

Sustainable Digital Architecture Framework

Executive Summary

Version 1.0 | September 2026 | Raymond du Toit

Sustainability becomes real through decisions

Sustainability commitments are ultimately delivered through ordinary decisions: what an organization designs, buys, builds, operates, modernizes, retains and retires. Those choices shape energy demand, embodied impacts, infrastructure utilization, asset life, supplier requirements, circularity and end-of-life outcomes. Many of their consequences persist for years and become costly to reverse.

The Sustainable Digital Architecture Framework, or SDAF, provides a structured way to integrate those consequences into consequential decisions. It connects lifecycle thinking, measurement, evidence quality, decision logic, circularity, governance, assurance and implementation within one coherent architecture.

Executive question	SDAF response
Why?	To make sustainability a controlled part of decisions rather than a consideration added after choices are substantially fixed.
What?	An open, vendor-neutral reference framework for evidence-based, transparent and defensible sustainability-related decisions.
Who?	Leaders and the architecture, sustainability, finance, procurement, engineering, risk, governance and assurance functions that support them.
Where?	Across projects, solutions, services, portfolios, enterprises, supply chains and ecosystems where choices create material lifecycle consequences.
When?	From strategy and design through procurement, operation, modernization, life extension, reuse and disposition - early enough to influence the outcome.

The central proposition

A sustainability claim is only as credible as the decision process behind it. SDAF strengthens that process by requiring comparable alternatives, explicit boundaries, traceable evidence, visible uncertainty, recorded trade-offs, appropriate authority and defined conditions for review.

It does not prescribe a preferred technology, replace specialist standards or guarantee a sustainable outcome. It establishes the decision architecture through which relevant evidence can be assembled, tested and used responsibly.

WHY AND WHAT

Why SDAF is needed

Organizations already use architecture methods, investment controls, procurement processes, operational telemetry, environmental accounting and risk governance. These mechanisms are valuable, but they often operate separately, use different boundaries and answer different questions. Their outputs do not automatically combine into a reproducible answer to a specific decision.

Recurring gap	Executive consequence	SDAF response
Evidence arrives after design choices narrow	Material impacts become locked in or expensive to change.	Orient establishes objectives, scope and requirements before evaluation.
Alternatives provide different functions	Comparisons can reward an option that does not meet the same need.	Baseline defines the Functional Unit, Reference Flow and system boundary.
Uncertain data appears precise	Decision confidence can exceed the strength of the evidence.	Evaluate records provenance, quality, uncertainty and sensitivity.
Trade-offs remain implicit	Leaders cannot see what was accepted, rejected or deferred.	Architect makes constraints, rationale, conditions and triggers explicit.
Approval substitutes for assurance	Governance may endorse a conclusion the evidence cannot support.	Assure separates decision authority from conformity and evidence review.

What SDAF is

SDAF is a vendor-neutral reference framework for incorporating sustainability into decisions with technical, financial, operational and lifecycle consequences. It provides common concepts, requirements, records and interfaces without mandating a particular organizational structure, product, supplier or calculation tool.

Its scope extends beyond carbon reporting. Depending on the decision, relevant evidence may include energy and water use, embodied impacts, material composition, repairability, utilization, service life, circularity pathways, cost, resilience, risk and social considerations. SDAF requires relevance and proportionality rather than the indiscriminate use of every possible metric.

What makes it different

- It starts with the decision and the function required, not with a preferred metric or technology.
- It distinguishes measured facts, modelled estimates, assumptions and professional judgement.
- It treats uncertainty, sufficiency and stability as decision controls rather than technical footnotes.
- It integrates lifecycle impacts and circularity while keeping them separately visible and traceable.
- It connects individual assessments to governance, assurance, organizational capability and external exchange.

WHO AND WHERE

Who SDAF is for

SDAF is designed for organizations in which responsibility for a consequential decision is distributed. It gives participants a shared structure while preserving their distinct authority and expertise.

Audience	Why SDAF is relevant
Executives and decision authorities	See the basis of a recommendation, its material trade-offs, evidence limitations, conditions and residual risks.
Architecture and engineering	Translate sustainability objectives into requirements, comparable options and design decisions.
Sustainability and data specialists	Connect environmental evidence to the decision context in which it can change an outcome.
Finance and procurement	Evaluate cost, value, supplier evidence, commercial constraints and lifecycle consequences together.
Risk, governance and assurance	Test conformity, evidence integrity, accountability, controls and review triggers without taking over the decision.
Standards bodies, educators and researchers	Use a common architecture for alignment, teaching, critique, extension and interoperability.

Where SDAF applies

SDAF can be used wherever a decision creates material environmental, technical, financial or operational consequences. It was developed with digital technology and IT estates in mind, but its architecture is intentionally general enough to be examined and adapted in other asset-intensive and decision-intensive domains.

Level	Illustrative application
Project or solution	Compare architecture options, deployment models, configurations or sourcing choices.
Service or product	Evaluate lifecycle impacts, demand, performance, resilience and end-of-use pathways.
Portfolio or enterprise	Align decision criteria, evidence requirements, investment governance and assurance across repeated choices.
Supply chain or ecosystem	Exchange comparable evidence, define supplier expectations and support interoperable records.

Authority is deliberately distributed

SDAF does not transfer all authority to a sustainability function. Executives retain decision authority; technical specialists retain professional responsibility; external laws and standards retain their own authority; and assurance remains independent of the decision it examines. The framework makes those boundaries explicit so that accountability is visible rather than assumed.

WHEN AND HOW

When SDAF should be used

SDAF should be applied early enough to affect the choice and maintained long enough to test whether its assumptions remain valid. The level of effort should be proportionate to the materiality, uncertainty, reversibility and consequence of the decision.

Decision stage	Relevant SDAF use
Strategy and investment planning	Define the objective, decision authority, hard constraints, material impacts and evidence expectations.
Architecture and solution design	Establish comparable functions, boundaries, scenarios and alternatives before committing to a design.
Procurement and sourcing	Specify evidence, data provenance, supplier obligations, service life and end-of-use requirements.
Implementation and operation	Confirm assumptions, monitor relevant performance and preserve the decision record.
Modernization and life extension	Reassess demand, utilization, replacement timing, reuse and alternative pathways.
Retirement and disposition	Evaluate data security, reuse, refurbishment, recovery and responsible treatment.
Assurance and review	Test conformity, evidence integrity, decision stability and the need to revisit the conclusion.

How SDAF works

Every SDAF assessment follows a controlled five-phase lifecycle:

Phase	Purpose
1 Orient	Define the decision, context, objectives, stakeholders, authority, requirements and material concerns.
2 Baseline	Establish the current state and a common basis for comparison, including the Functional Unit and Reference Flow.
3 Evaluate	Generate, normalize and test the evidence, including its provenance, quality, uncertainty and sensitivity.
4 Architect	Apply constraints and decision tests, assess trade-offs, select or defer an option and record the rationale.
5 Assure	Confirm conformity and evidence integrity, document findings and define monitoring or reassessment conditions.

Eight Parts provide the requirements that support this lifecycle: Foundations; Lifecycle; Measurement; Decision-Making; Lifecycle and Circularity; Governance and Assurance; Enterprise Operationalization; and Ecosystem Interoperability.

VALUE AND NEXT STEPS

What value SDAF can provide

SDAF is valuable when it improves a decision, not when it merely produces more documentation. Its value should therefore be judged through the quality, transparency and durability of the decisions it supports.

Potential value	What it means in practice
Better decision quality	Relevant lifecycle consequences and material trade-offs are considered before commitment.
Greater defensibility	The evidence, assumptions, rationale, authority and conditions behind a conclusion can be examined.
Proportionate confidence	The strength of the conclusion reflects the quality and completeness of the available evidence.
Clearer accountability	Decision, methodological, data, governance and assurance responsibilities are distinguishable.
Reduced lock-in and rework	Consequences are surfaced while alternatives remain open and corrective action is still practical.
Organizational learning	Findings and operating evidence inform later decisions and improve methods, data and capability.
Interoperability	Controlled records can align with external standards, supplier information and related frameworks without surrendering traceability.

How an organization can engage

- Explore the full framework to understand its concepts, requirements and authority boundaries.
- Map existing architecture, sustainability, procurement and governance practices against SDAF to identify alignment and gaps.
- Apply the relevant requirements to a consequential decision, portfolio or operating process at a proportionate depth.
- Review SDAF from an institutional, standards, assurance, educational or research perspective and provide informed critique.
- Consider where SDAF could complement, connect or strengthen established methods rather than duplicate them.

An open invitation

SDAF v1.0 is published openly to encourage use, scrutiny and improvement. Organizations and practitioners are invited to examine whether it helps translate sustainability ambition into better decisions, to adapt their engagement to their context, and to contribute evidence and experience to its continued development.

The immediate next step is to examine the complete framework and determine whether its architecture can strengthen the decisions for which your organization is accountable.

Full framework: <https://doi.org/10.5281/zenodo.22899663>

Review and contribution: <https://github.com/Raymond-du-Toit/SDAF/blob/main/CONTRIBUTING.md>

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